Advancing Genomic Data Interoperability: The FEGA Metadata Model

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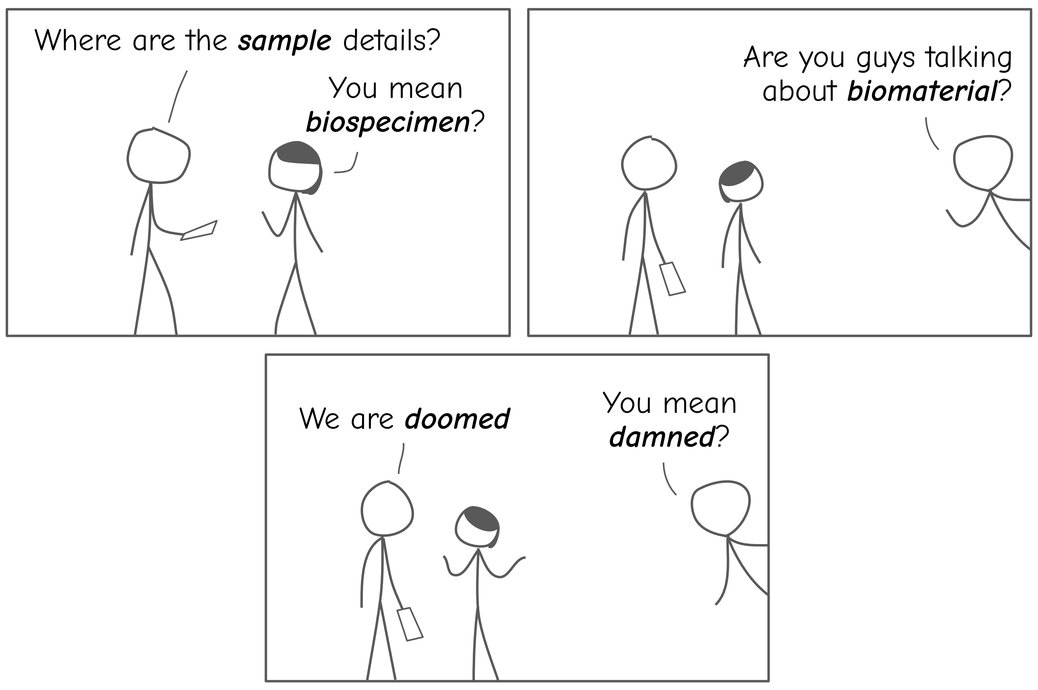
The **Federated European Genome-phenome Archive** (FEGA)[1] is a collaborative network designed to enhance global discovery and controlled access to sensitive human omics data. We face the challenge of harmonising heterogeneous data models, whose diversity poses a significant barrier to achieving true interoperability among data archives.

In response, the FEGA Metadata Working Group is set to introduce the FEGA Abstract Metadata Model. Our approach emphasises the creation of a Resource Description Framework (**RDF)-compliant abstract model**[2] based on existing models from the FEGA network, as well as known and emerging standards.

It not only facilitates data sharing and utilisation within FEGA nodes but also establishes **alignment with broader initiatives** like the **European Genomic Data Infrastructure** (GDI)[3] project, the **Global Alliance for Genomics and Health** (GA4GH)[4], and **1+ Million Genomes** (1+MG)[5] **initiative**.

The *Telephone Game* of the adult world



Survey Existing FEGA Nodes
Collected metadata requirements from FEGA nodes and GDI

Feedback phase
Gathering feedback and refining the abstract model

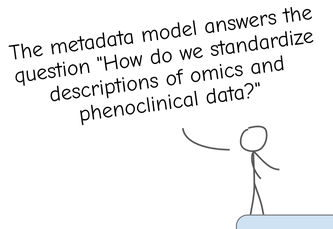
Serialize Model
Converting the abstract model into a machine-readable format with validation (e.g., JSON Schema)[6]

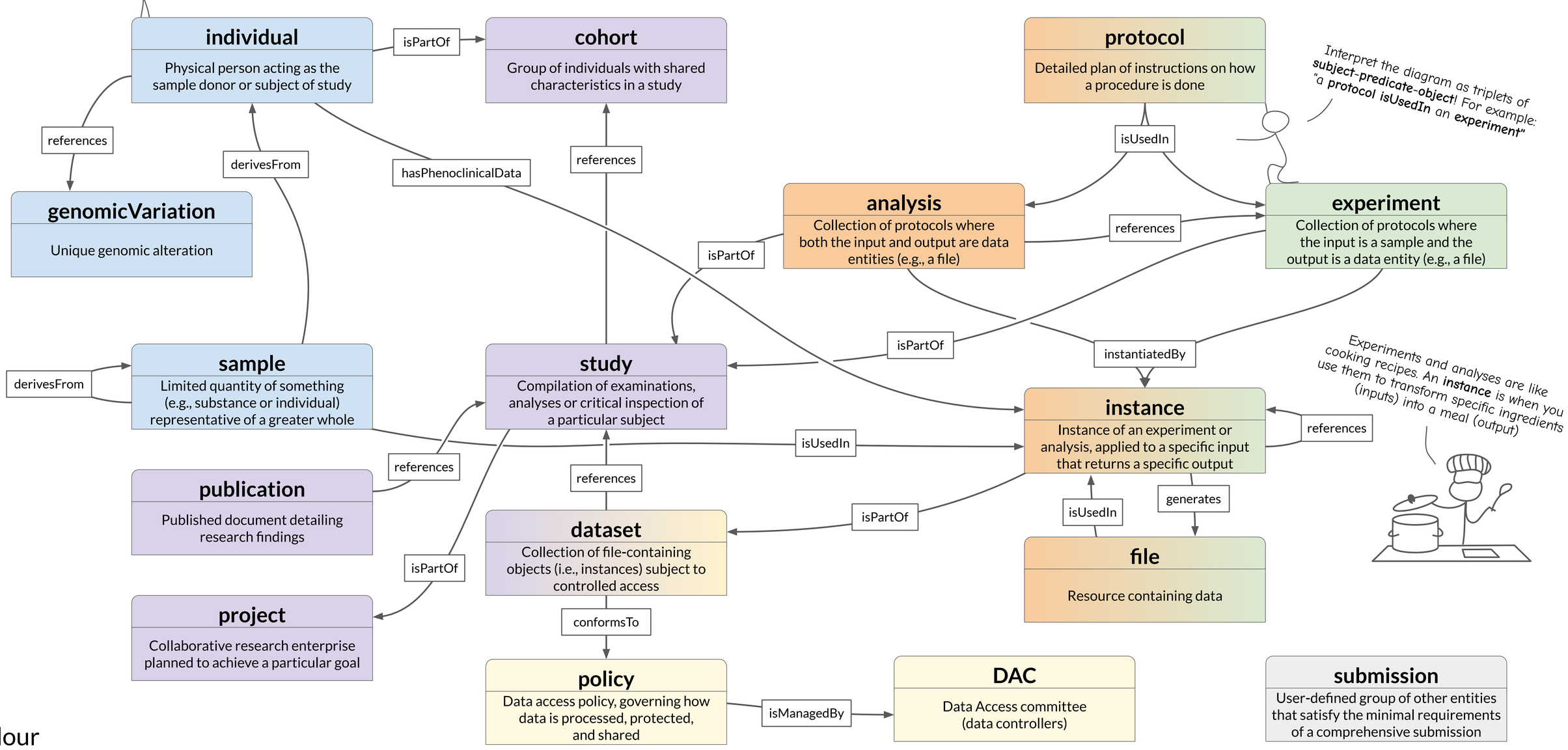
January 2024 — March 2024 — May 2024 — July 2024 — Q4 2024 — Q2 2025

Draft Combined Abstract Model
Created an initial draft of the unified metadata model

Finalize Abstract Model
Completing and finalizing the abstract metadata model

Evaluate Compatibility
Assessing the model's compatibility (FEGA nodes, Beacon Network[7], GDI...)





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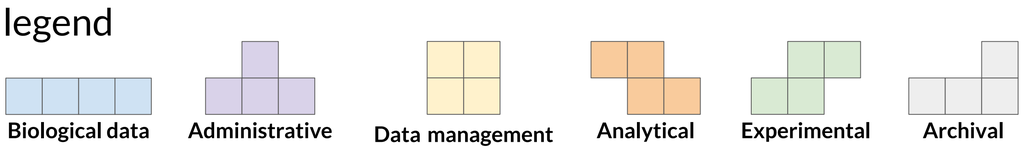

This is a **collaborative effort**! The main author thanks the members of the FEGA Metadata Working Group for their valuable contributions and feedback.

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